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Digital teaching resource management system for higher education

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Abstract

At present, the management and utilization of teaching resources in higher education are characterized by issues, such as limited functionality, simple structure, inadequate sharing, and weak management capacity. To address these challenges, this study proposes a multimedia information processing technique based on computer technology. The system utilizes Java programming language, the Spring Boot framework, Servlet, and other technologies to process data through a MySQL database. It consists of main modules for students, teachers, and administrators. The developed digital teaching resource management system is stable, user-friendly, with a clear interface, and capable of meeting the educational needs of colleges and universities. By maximizing the sharing and integration of university teaching resources, this system not only innovates teaching and learning approaches but also holds significant practical implications.

KEYWORDS

database, digitization, teaching resource

1 | INTRODUCTION

As the mainstream media, the Internet is increasingly used in the field of education, and students and teachers are increasingly dependent on computers in their daily lives.[1–3] According to incomplete data, every student spends at least 4 h browsing news and learning about their expertise online.[4] Therefore, if we can put our educational resources online and rely on the efficiency and convenience of the Internet, we will be able to do a better job of learning from digital educational resources.

This paper proposes a multimedia information processing technology based on computer technology and establishes a digital teaching resource management system (DTRMS) for universities. In DTRM in universities, the goal is to change the existing manual or semi-manual university DTRM mode, so that the DTRMS becomes more standardized and regulated, thus enhancing work efficiency.

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The system utilizes Java programming language, Spring Boot framework, Servlet and other technologies[5,6] to convert data information through MySQL database. It mainly implements the functions of homepage, personal center, student management, teacher management, resource classification, and management of the TR system (cyclic charts, notification information, about us, system introduction). The system builds a more scientific and efficient DTRM process for students and teachers, thus reducing the errors that exist in traditional paper-based university TRs. The system has a simple and clear operation interface, which is easy for users to understand quickly, thus greatly improving the efficiency of software development.

The system uses network communication, computer storage and management of information,[7–9] which has irreplaceable advantages of traditional methods. For example, it is very fast, very reliable and has a very large storage space. In particular, it has excellent confidentiality, very long retention time, and very low cost. In terms of work efficiency, it can be significantly improved, expanding to a higher level of service will pay good dividends, combined with the Internet.

The system integrates rich digital resources and uses data analysis and statistical functions to track students' learning progress and effectiveness in real time, providing accurate teaching feedback for teachers. At the same time, the statistical analysis of the use of teaching resources can provide strong support for school resource allocation and decision-making. In addition, the system can optimize resource allocation, improve teaching efficiency and quality, and promote innovation and development of educational models.

The established system has significantly enriched literary education by integrating, optimizing, and sharing diverse resources. It provides a wide range of materials, such as e-books, audiovisual resources, and multimedia courseware. This enhances teaching quality by offering efficient access to necessary resources, replacing traditional methods. Moreover, it promotes international literary development, exposing students to diverse cultural perspectives, thereby broadening their global outlook.

2 | DEVELOPMENT TOOLS AND RELATED TECHNOLOGY

2.1 | Spring Boot framework

Spring Boot follows a layered architecture (Figure 1), where each layer communicates with its direct lower or upper layer (hierarchy). Spring Boot has display layer, business layer, persistence layer, and database layer.[10–12]

1. Presentation layer: Responsible for processing HTTP requests, converting JSON parameters into objects, validating requests, and forwarding them to the business layer. In essence, it encompasses the view (i.e., the front end).
2. Business layer: Handles all business logic, comprising service classes that utilize functionalities provided by the data access layer. It also manages authorization and authentication.
3. Presentation layer: Contains all storage logic and facilitates the conversion of business objects to and from database rows.
4. Database layer: Primarily responsible for CRUD operations (Create, Read, Update, Delete).

2.2 | Vue framework

Vue is a JavaScript framework used to build user interfaces.[13–15] It builds on standard HTML, CSS, and JavaScript to provide a component-based, declarative programming model for more efficient user interface development. Vue is capable of handling both simple and complex interfaces. Vue has two core functions:

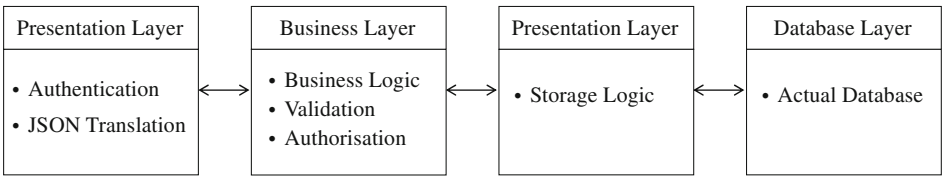


FIGURE 1 Spring Boot structure.

1. Declarative rendering: Vue has extended a template syntax based on standard HTML, allowing us to declaratively describe the relationship between the final output HTML and JavaScript state.
2. Responsiveness: Vue automatically tracks the JavaScript state and updates the DOM in a responsive manner when it changes.

2.3 | MySQL database

MySQL is a relational database management system.[16,17] Relational databases store data in different tables rather than keeping all the data in one large repository, which increases speed and improves flexibility. The SQL language used by MySQL is the most commonly used standardized language for accessing databases. The MySQL software adopts a dual-licensing policy and is divided into a community edition and a commercial edition. Due to its small size, high speed, low total cost of ownership, especially open source this feature, generally small, medium and large website development choose MySQL as a website database. MySQL can be seen as a three-layer structure:

1. The first layer Connection Layer is responsible for handling connections and disconnections with clients, as well as security mechanisms, such as authentication and authorization. At this layer, MySQL will convert client requests into internal formats and pass them to the SQL or service layer. The connection layer is also responsible for managing connection pools to improve connection efficiency and performance.
2. The second layer of SQL Parser Layer is where all work before processing the underlying data in the MySQL database system is completed, including permission determination, SQL parsing, execution plan optimization, query cache processing, and so on.
3. The third layer of Storage Engine Layer, which is the implementation part of the underlying data access operations. It is composed of multiple storage engines. The basic structure of MySQL is shown in Figure 2.

2.4 | Ajax technology

Ajax(Asynchronous Javascript And XML)includes HTML or XHTML, CSS, JavaScript, DOM, XML, XSLT, and XMLHttpRequest.[18,19] Using Ajax technology, web applications can quickly present incremental updates to the student interface without having to reload (refresh) the entire page, which allows the program to respond more quickly to the student's actions.

The biggest advantage of Ajax is that it can exchange data with the server and update some webpage content without reloading the entire page. The principle basis of Ajax implementation is that web DOM objects can accurately manipulate certain parts of the content in the web page. As a simple data storage carrier, XML can enable clients and servers to exchange only web page content data without accompanying information such as web page styles. XMLHttpRequest is a request object that interacts with the server independently of the browser's built-in request. The operating principle of Ajax is shown in Figure 3.

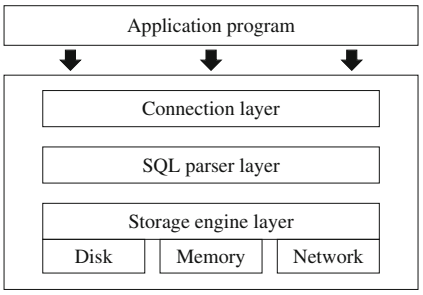


FIGURE 2 MySQL basic structure.

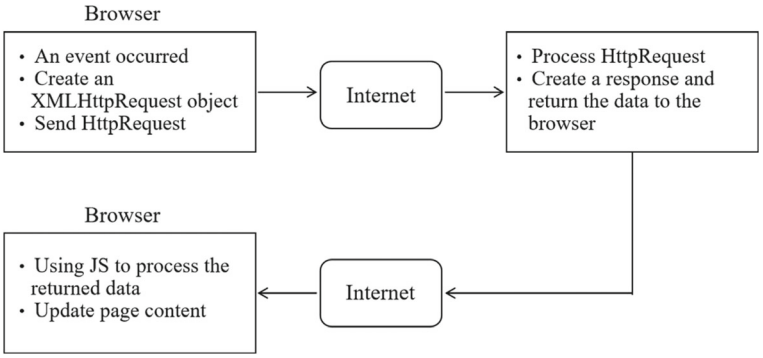


FIGURE 3 Ajax operating principle.

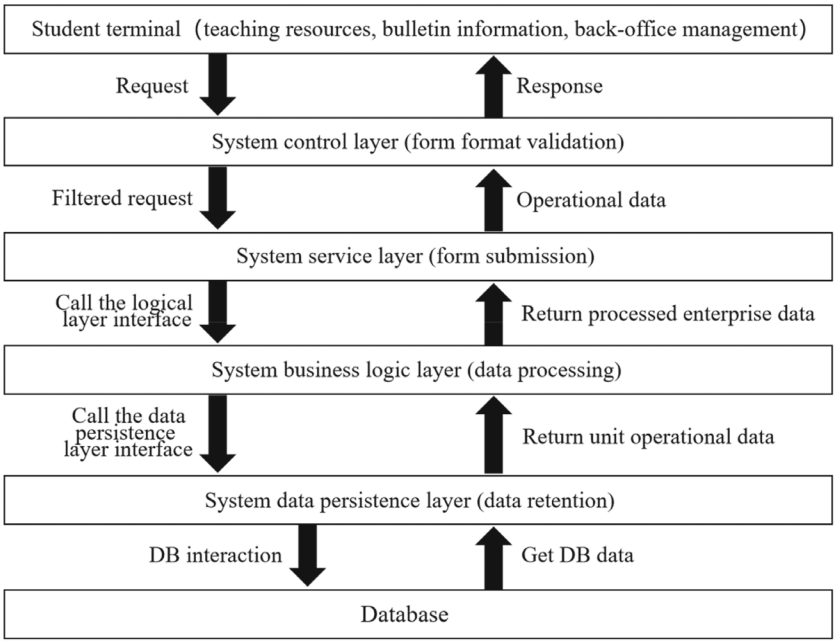


FIGURE 4 System framework diagram.

3 | DTRMS FOR UNIVERSITIES

3.1 | System framework

This system is designed from five levels, including presentation layer, control layer, service layer, business logic layer, data access layer.

The main function of a presentation layer is to interact with students and provide them with excellent learning experience. The system adopts Ajax technology, making it run smoothly without the need for updates.

Control layer is the real-time processing of data from the front-end and back-end interfaces, and adjusts the interfaces accordingly according to the requirements of the front-end interface.

Service layer is an interface between the business logic layer and the control layer, where various information services in the business logic layer are recombined and transmitted as data to the business control layer.

Business logic layer refers to the dynamic invocation of methods in the data access layer, the implementation of complex algorithms, the processing of requests sent from the front-end, and the corresponding logical processing.

The data access layer is used to access databases, text files, binary files and XML formatted documents. Manage the addition, modification, removal and loading of business objectives. The system framework diagram is shown in Figure 4.

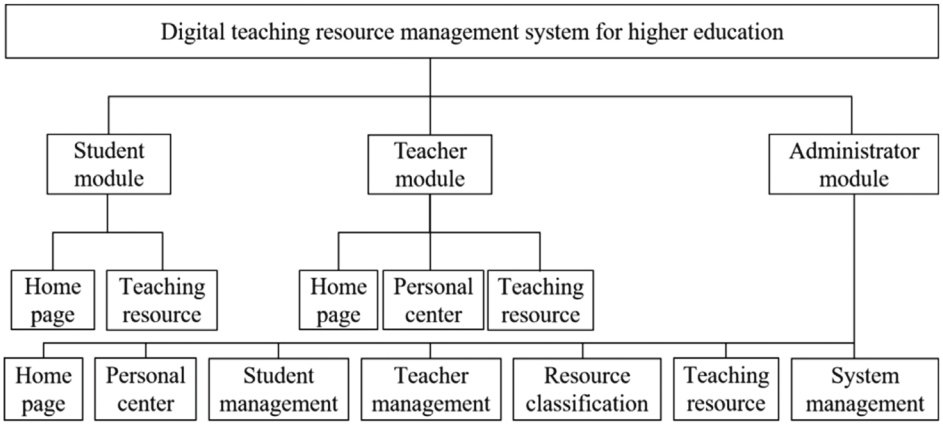


FIGURE 5 System functional structure diagram.

3.2 | Functional module

The basic functional requirements of the system have been obtained through software requirement analysis. According to the different functional modules, the system is divided into various functional blocks (Figure 5).

The front desk mainly completes functions, such as student login and registration, display, query, and collection of TRs.

1. Learning materials: this website contains a large amount of learning materials that can be freely read and selected from which content is of interest.
2. Search for TRs: determine clear goals and search for the required content in the search box, which can help you find TRs faster and more accurately.
3. Announcement information: view the information and system introduction published by the system.

The backend of the system includes: student management, teacher management, TRM and resource classification.

1. Student management: query, modify and delete the information of all registered students.
2. Teacher management: query, modify and delete registered teachers.
3. Resource classification management: administrators can add and delete resource classification.
4. Processing of teaching materials: teachers can add, modify and delete teaching materials. Administrators can modify and delete TRs.
5. Add and delete the figures.
6. Notification information: administrators can add, modify and delete information.

3.3 | Database design

3.3.1 | Conceptual model design

Conceptual model design is based on requirement analysis, analyzing requirements and abstracting them into concepts. This approach allows developers to simplify complex problems and focus on dealing with patterns and organizational structures. The most common among these theories is the E-R mode.[20–22] The system E-R diagram is shown in Figure 6.

Figure 6 shows the browsing of TRs by students. The updated TRs by the teacher include TR numbers, names, classifications, and so forth, with a relationship of 1: N. The relationship between images, documents, and so forth. is 1: N. The relationship between the TRs to be learned is 1:1.

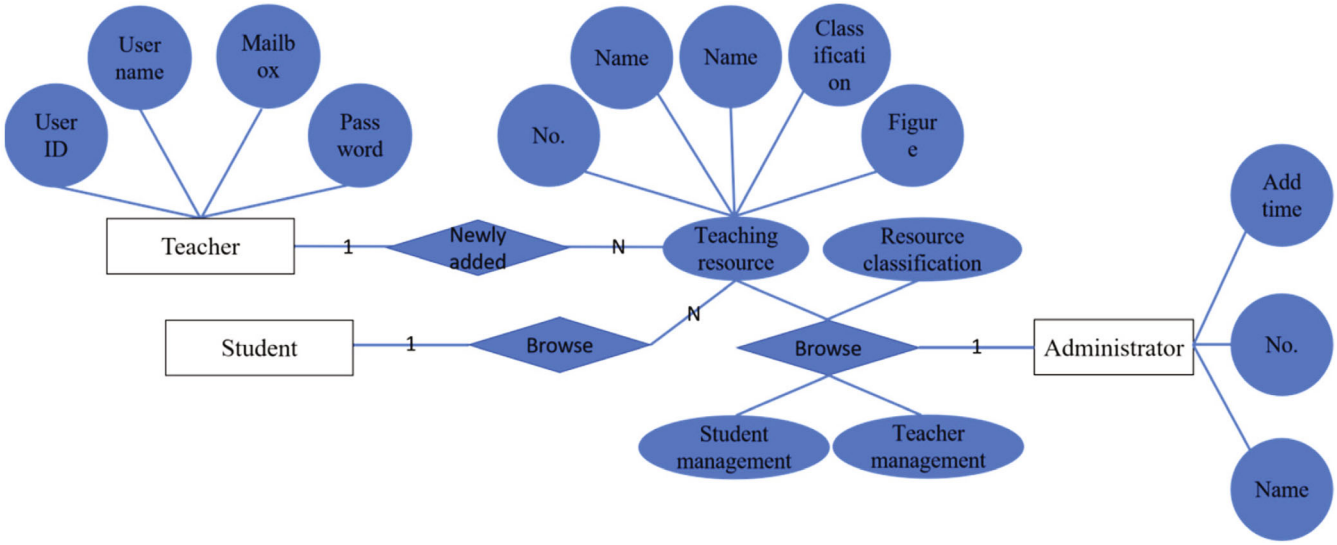


FIGURE 6 E-R diagram of the system.

3.3.2 | Database logic design

Based on the entities and attributes of the system, combined with data design concepts, the logical design of the system is described as follows.

1. Personal center (number, password modification, personal information).
2. Student management (ID, student ID, name, profile picture, password, gender, phone number).
3. Teacher management (ID, job number, name, profile picture, password, gender, phone number).
4. Resource classification management (numbering, new resource classification).
5. TRM (number, name, category, image, document, job number, name, publication time, click count).
6. Banner figure's management (number, name, image).
7. Announcement information (number, title, introduction, content, image).

4 | SYSTEM IMPLEMENTATION

4.1 | Implementation of student login

Students log in to the webpage, which has different permissions and distinguishes administrators from users. Log in to the account of the system administrator to manage TRs in the background. Ordinary users can operate the client, register and log in, and browse TRs on the network. If the user has not yet logged in with their account and password, they must first register in this system. After logging in, you can view and modify your personal information and password, and access relevant resources. In the window of this system, students can access their personal information and change their passwords.

4.2 | Main function of the system front-end

4.2.1 | Implementation of homepage

The operation interface should be as simple and clear as possible, making it easy for students to search for functional entrances and browse TRs. Simplify interface changes and maintenance steps to ensure the legitimate rights and interests of students and system security.

4.2.2 | Implementation of student registration

The student registration webpage needs to fill in the name, password, name, address, email, and so forth, and these information cannot be blank. Users can change relevant configuration files and passwords by clicking the button. For registered users, the system administrator will verify their identity, and those who fail will cancel their login. In summary, as long as it is not deleted by the administrator, it is considered a formal student. Users can only log in to the system after completing registration. This module function is executed in web pages.

4.2.3 | Implementation of TR display

The TR display page is a page that displays the TRs of DTRM. It arranges all TRs in descending order by number using SQL statements, such as "select shangpinxinmingcheng, id, tupian, jiage, addtime from shangpinxi order by addtime desc".

4.2.4 | Implementation of TR search

The TR search engine provides users with a search for TRs, making it easy for users to find the resources they need. Select an option from the user's dropdown list, enter relevant information, such as TR name, TR status, and publisher, and click OK to search for the corresponding TR information.

4.2.5 | Implementation of browsing TR

The shopping cart page is mainly used for trading TRs. After the user logs in, search for the required TRs and add the selected TRs to the shopping cart. After the order is completed, you can view the brief introduction of the relevant resources.

4.3 | Implementation of system function

4.3.1 | Implementation of student management

Administrators manage students in the system, including entering, deleting and modifying them. Obtain the student's name through SESSION, then enter the new password and submit it to mod.html. Use SQL commands to update the password. The student management interface is shown in Figure 7.

4.3.2 | Implementation of DTRM

After logging in to the account on the homepage, the administrator can perform backend management. Searching for TR information through numbers, photos, prices, and so forth has significant advantages in the classification of TRs. In the submodule of TR directory information management, system administrators can modify TR directory information, achieving support for teacher education and scientific research. Among them, the name, introduction, types, and prices of learning materials have been added to the file, and photos and teaching materials have been provided. The search interface for TR in DTRM system is shown in Figure 8.

4.3.3 | Implementation of DTRM

The administrator clicks on the all DTRM interface dingdan_list.html through the backend management interface, and then requests the control layer dingdanAction to query the information of all current DTRs. DingdanAction calls the request to search for all DTRs from dingdanDAOImp. In the DTR table of the database, after searching for all the current

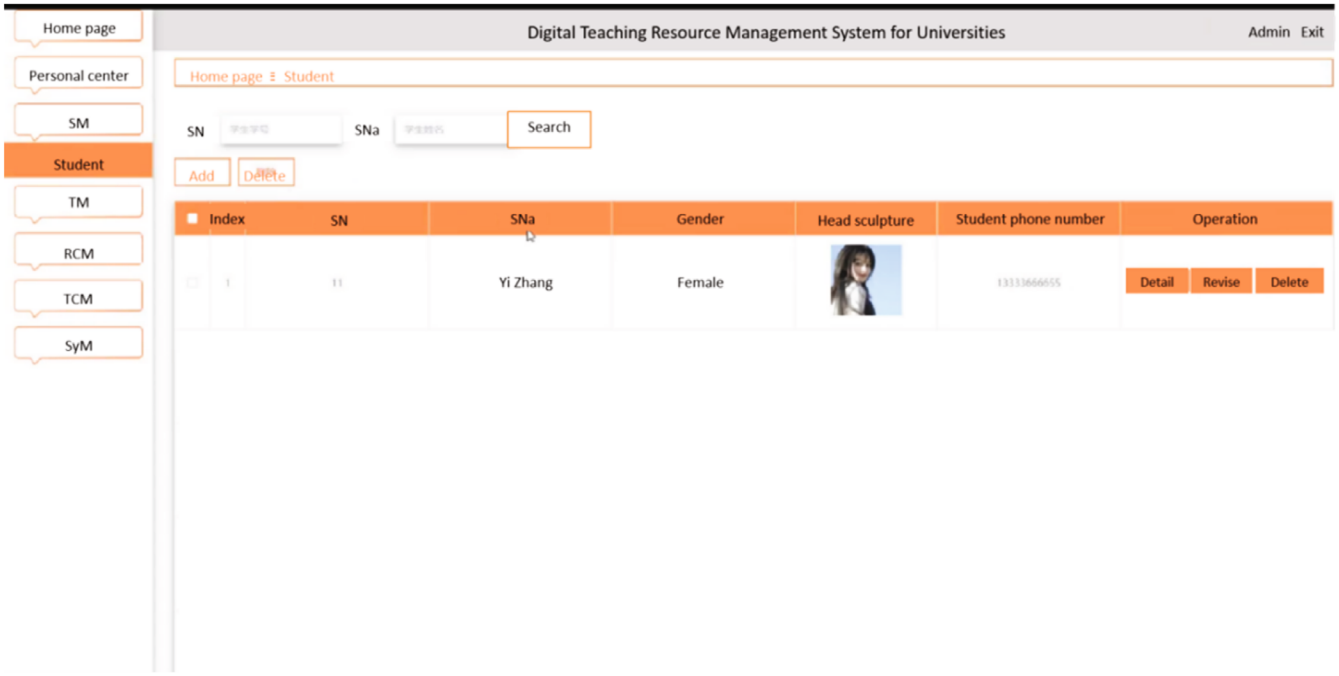


FIGURE 7 Student management interface.

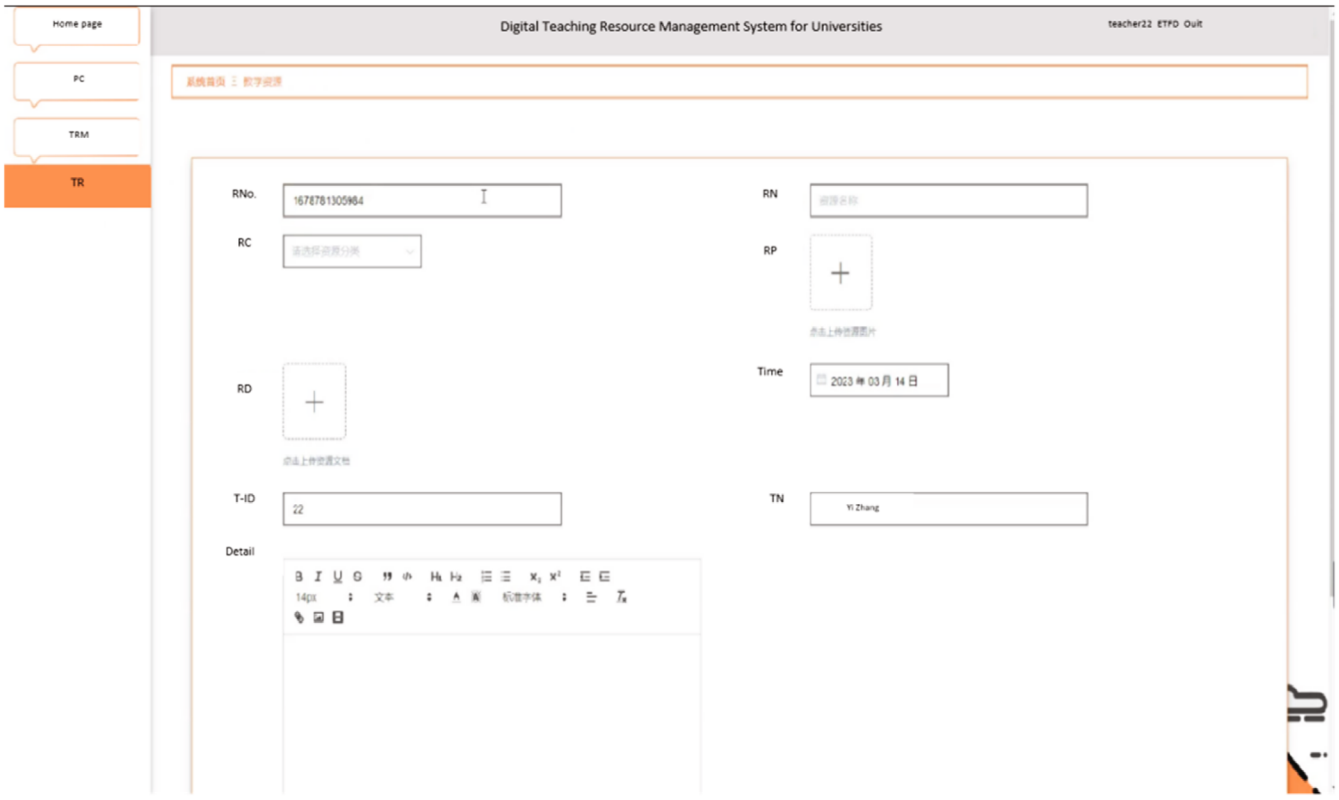


FIGURE 8 Search interface for TR in DTRM system.

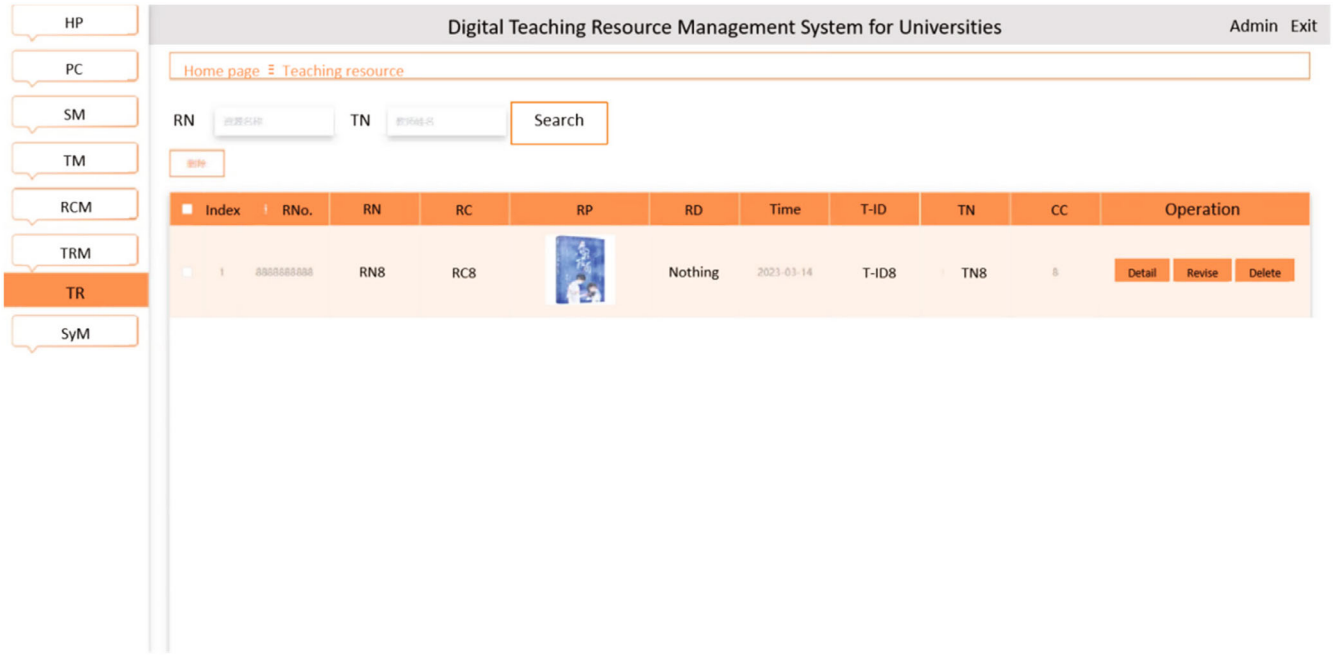


FIGURE 9 Example of searched TR interface.

DTRs, return layer by layer to the dingdan_list.html interface to display the information of all the current DTRs. After deleting expired digital education resources on this interface, existing digital education resources can be verified before being sent out. An example of DTR management interface is shown in Figure 9.

5 | SYSTEM TESTING

5.1 | Test environment construction

1. Hardware environment

The server hardware environment is configured with a 2.0 GHz quad core CPU, 4G DDR3 or higher memory, 2 TB hard disk, and 512 M independent graphics card.

The client hardware environment is configured with a 1.8 GHz dual core CPU, 1 GB or more of memory, 500 GB or more of hard disk, and a 512 MB or more independent graphics card.

Network equipment: 100 M or above network adapter, network card, network cable.

2. Software environment

Server software environment: the operating system adopts Windows Scrvcr 2003, the database management system adopts SQL Server 2008 Enterprise Edition, and the server installation application software is the university DTRMS.

Client software environment: the operating system adopts Win 10 and E11.0 and above browsers are installed.

According to business needs, 100,000 pieces of data are deployed in the database. The devices used for testing are all currently popular hardware configurations, and then the university TRMS is tested.

5.2 | Functional testing

To ensure the stable operation of the university DTRMS, this study employs black-box and white-box testing methods integrated with its core functionalities, specifically resource service and management. These tests are conducted to verify the system's reliability and accuracy in performance.

Following a thorough analysis involving at least five tests per module, the study concludes:

1. The system successfully implements user login and authentication.
2. It supports resource uploading and keyword-based resource searches.
3. System administrators can manage user permissions and conduct resource audits effectively.

Post-testing, the university DTRMS interface is simple, user-friendly, meeting expectations, and fulfilling user needs.

5.3 | Performance testing

Performance testing focuses on three main indicators: load testing, stress testing, and capacity testing. Load testing evaluates system performance under expected loads, ensuring monitored values align with expectations. Stress testing applies continuous pressure to assess system operation. Capacity testing addresses issues related to managing large data volumes, verifying compliance with testing requirements.

The objectives of this performance test include identifying system performance bottlenecks in the current environment and determining the maximum number of concurrent users the system can support while maintaining acceptable response times under specified data volumes.

Compuware's QALoad serves as the performance testing tool, simulating multiple concurrent user requests and capturing system throughput and response time data. Testing occurs within Microsoft's Windows 2003 and SQL Server 2000 (SP3) environments. Results indicate stable software performance, supporting deployment of a system environment.

5.4 | Functional testing

This software performs related functions, such as querying, adding, deleting, and modifying during operation. The displayed result is normal. The queries of the backend management database are also running normally, ensuring the consistency and reliability of data information.

After a complete testing process, all functions of the system are running stably and all interfaces are displaying normally. After the product is correctly uploaded to the mall system, users can run the purchase function correctly, the administrator can accurately manage and operate related orders. The specific tests are shown in Table 1.

5.5 | Implementation effects

To better understand the current usage, satisfaction, and areas for improvement of the university's DTRMS, a comprehensive survey (Table 2) is conducted among faculty and students from diverse backgrounds. The survey covered usability, content quality, accessibility, interaction, feedback mechanisms, technical support, and assistance. Using online methods and data analysis tools, the collected feedback is carefully categorized and analyzed.

Based on the analysis of questionnaire results, usability evaluations showed that most users positively rated the system's interface design, though some found certain operations confusing. In terms of resource content and quality, the resource variety was well-received, but users demanded faster updates and better quality control. For accessibility and compatibility, most users accessed the system smoothly, but a few reported issues on specific devices or networks. Regarding interaction and feedback mechanisms, users wanted more interactive features, such as online discussion forums and teacher Q&A, along with more timely and effective feedback. Technical support received moderate satisfaction, but users highlighted the need for faster response times and efficiency in resolving urgent issues.

To address these points, the following actions are recommended:

1. Optimize the interface and simplify operation processes to reduce learning costs and enhance user experience.
2. Implement stricter resource review processes to ensure timely and high-quality content, and broaden resource sources by collaborating with publishers and academic institutions.
3. Test and optimize the system for various devices and network environments to ensure stable operation and minimize compatibility issues.

TABLE 1 System functional test table.

Number	Test function	Test content	Test result
1	Student Login	1. Verify that the student name and password are correct. 2. Verify that the password is visible.	Pass
2	Home display	1. Whether the home page data is loaded successfully. 2. Verify the accuracy of the search function. 3. Verify that it can be loaded asynchronously. 4. Verify the navigation bar button.	Pass
3	Personal information modification	1. Verify that the login name can be changed properly. 2. Verify that the contact information can be changed. 3. Verify that the shipping address can be changed properly. 4. Verify that the password can be changed.	Pass
4	Resource classification management	1. Verify that the category addition can be successful. 2. Verify that the category deletion can be successful.	Pass
5	TRM	1. TR information is consistent with the upload. 2. Whether or not the price modification can be completed. 3. Verify inventory information changes.	Pass
6	Add TR	1. Whether to add validation for uploading TR. 2. Whether the picture is uploaded successfully. 3. Verify that the form is submitted successfully.	Pass
7	Student management	1. Verify the student entry function. 2. Verify the student violation cleanup function.	Pass

TABLE 2 User experience survey questionnaire for DTRMS.

Item	Option
1. Interface design	☐Very satisfied ☐Satisfied ☐Generally ☐Dissatisfied ☐Very dissatisfied
2. Resource content and quality	☐Very satisfied ☐Satisfied ☐Generally ☐Dissatisfied ☐Very dissatisfied
3. Functional practicality	☐Very satisfied ☐Satisfied ☐Generally ☐Dissatisfied ☐Very dissatisfied
4. Access and compatibility	☐Very satisfied ☐Satisfied ☐Generally ☐Dissatisfied ☐Very dissatisfied
5. System stability	☐Very satisfied ☐Satisfied ☐Generally ☐Dissatisfied ☐Very dissatisfied
6. Usability	☐Very satisfied ☐Satisfied ☐Generally ☐Dissatisfied ☐Very dissatisfied
7. Interaction and feedback	☐Very satisfied ☐Satisfied ☐Generally ☐Dissatisfied ☐Very dissatisfied
8. System help document	☐Very satisfied ☐Satisfied ☐Generally ☐Dissatisfied ☐Very dissatisfied
9. Overall satisfaction	☐Very satisfied ☐Satisfied ☐Generally ☐Dissatisfied ☐Very dissatisfied

4. Add interactive features like online discussion forums and teacher Q&A to enhance user interaction and collaboration.
5. Improve feedback mechanisms to ensure timely responses and handling of user issues and suggestions.
6. Enhance the technical support team's expertise and service levels to quickly resolve user problems.
7. Conduct regular user training sessions to help users master system usage techniques.

6 | DISCUSSION

The entire development process primarily focuses on system functional modules, including requirement analysis, system design, and implementation. The digitalization of teaching resource management systems in universities not only achieves resource sharing but also introduces a new teaching approach, effectively enhancing learning efficiency.[23,24]

In-depth discussions on the construction and operation of university teaching resource management systems revolve around a series of research hypotheses or expected goals. These hypotheses not only guide the system's development direction but also gradually validate their feasibility and rationality through practical application feedback and adjustments.[25,26]

First, it is hypothesized that an efficient teaching resource management system can significantly improve teaching quality and efficiency. Thus, the system integrates various educational resources, such as e-books, online courses, teaching videos, and lab instructions, providing a one-stop learning and teaching platform for students and teachers. Data analysis shows increased student usage frequency, duration, and improved learning outcomes, objectively verifying this hypothesis.

Second, the system is expected to optimize resource allocation and reduce waste. Through big data analysis and intelligent recommendation technologies, the teaching resource management system accurately predicts faculty and student needs, facilitating resource allocation. Comparative analysis of resource utilization rates and cost inputs before and after implementation clearly demonstrates the system's positive impact in this regard.

Third, it is hypothesized that the system can enhance interaction between faculty and students, fostering mutual learning. Built-in features like discussion forums and online Q&A platforms provide a convenient communication platform, ensuring timely resolution of queries during the learning process and promoting intellectual exchange and knowledge sharing among faculty and students. Data collected on faculty-student interactions indicates increased participation and satisfaction levels, further validating the system's value in this aspect.

In conclusion, the university teaching resource management system, through hypothesis-based research and objective analysis using facts, data, and evidence, underscores its significance. These analyses not only provide robust support for system development but also guide future optimization and improvements.[27]

This paper adopts an interdisciplinary perspective and diverse methodological applications, reiterating the core value of teaching resource management systems and addressing key challenges, while offering unique insights and innovations:

1. Innovatively introduces User Experience Design (UXD) theory. This paper combines in-depth interviews, surveys, and big data analysis to delve into user behavior patterns and preferences, offering more personalized and user-centric solutions for optimizing teaching resource management systems.
2. Integrates social science theories into system design. Through emotional design and incentive mechanisms, it stimulates faculty and student learning motivation and creativity, facilitating effective flow and sharing of teaching resources.
3. Innovates new models for teaching resource management. It proposes a holistic 'Technology-Humanities-Management' approach to teaching resource management, emphasizing that technological innovations should serve humanistic care and educational goals, while focusing on organizational change and institutional safeguards during system implementation.

In summary, this paper critically inherits and innovatively develops previous research achievements, revealing new trends and directions in university teaching resource management system research, demonstrating its unique value and application potential in practice. This study not only enriches theoretical contributions to the academic community but also provides valuable references and insights for educational practitioners.

7 | CONCLUSION

The TR system facilitates TR sharing, enhancing learning efficiency and fostering teaching development. Establishing a network resource management system proves the most convenient and effective sharing method. This study focuses on constructing a DTRMS tailored for universities. Through requirement analysis, framework exploration, and detailed development using advanced design concepts and mature network technologies, the system was developed. Implementation of the university DTRMS achieves the following objectives:

1. Easy maintenance: Enables real-time user management and content processing, ensuring quick front-end and back-end interaction.

2. Simplified data statistics: With minimal tables, primarily user and information tables, the system ensures straightforward data maintenance.
3. Streamlined interface: Offers fast, flexible navigation with simple operations, reducing system coupling. It separates dynamic page control from static content display, ensuring unified styling and layout.
4. Efficient query and retrieval: Provides real-time query capabilities tailored to the needs of teachers and students at different levels.

Developing a university TR information management system is a comprehensive initiative aimed at maximizing TR integration and promoting information sharing across the university. The benefits brought to universities mainly include the following aspects.

1. Improve the quality of teaching. By integrating diverse teaching resources such as lesson plans, courseware, instructional videos, and so forth, the proposed system provides teachers and students with more diverse learning paths and methods.
2. Improve teaching efficiency. The system supports rapid sharing and communication of teaching resources, making it easier for teachers and students to access the necessary resources, saving time and effort in searching and organizing resources.
3. Reduce the cost of education. By unifying and integrating teaching resources, this system can avoid duplicate development and purchase of resources, reducing the school's investment in teaching resources.
4. Promote teaching innovation. This study provides teachers with diverse teaching methods and tools, such as blended learning, flipped classrooms, and so forth, encouraging teachers to engage in teaching innovation and practice.
5. Promote the overall development of the university. This system can integrate existing resources, such as teaching, curriculum, and employment in schools to form a unified resource pool for easy management and archiving.

The DTRMS has undergone stable testing and operation, proving user-friendly and meeting the teaching and resource-sharing needs of universities. Its implementation enhances the scientific management of TRM, reduces labor intensity, and improves efficiency, offering significant practical benefits.

Following in-depth research, design, and development using the Spring Boot framework, the university DTRMS has been largely completed. It currently manages students, teachers, and resource classifications. However, time constraints and knowledge gaps have resulted in some deficiencies. For instance, the system lacks a payment module integrated with banking systems, secure access constraints, and school DTR reporting and data output capabilities, necessitating further refinement.

During the university TRMS construction, we have identified the need for enhanced computer knowledge to overcome technical challenges effectively. Information technology advancement is increasingly integral to modern education, making the construction of DTRMS essential for enhancing teaching quality, methods, and overall institutional strength. Future advancements in information technology will continue to drive educational reforms, significantly advancing the education sector.

Our goal is to leverage information system construction to enhance school management efficiency and facilitate data sharing among university departments. Implementing an integrated information management system aims to enhance networking and optimize TR utilization across the institution.

AUTHOR CONTRIBUTIONS

Lin Zhu: Writing – original draft; methodology; data curation; formal analysis. **Li Zhuang:** Validation; writing – review and editing; investigation; conceptualization; funding acquisition. **Caihong Li:** Data curation; formal analysis.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflict of interest.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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